

Ilia Karmanov

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Overview

Senior Staff Research Scientist at NVIDIA ADLR, working on vision-language models across pre-training, post-training, and evaluation for multimodal reasoning. First author of Eclair, a document-understanding model adopted across NVIDIA's pipelines for pre-training data extraction and VLM distillation, and shipped as Nemotron Parse. Earlier, published at NeurIPS, ICCV, and BMVC at Qualcomm AI Research. Trained in economics at LSE (BSc First Class; MSc), with a focus on optimal control and causal inference, before moving into machine learning in 2016.

Experience

NVIDIA ADLR, Zurich

Mar 2025 – present

Senior Staff Research Scientist

- Contribute across NVIDIA's Nemotron vision-language models (Nano V2 VL, Nano 3 Omni): pre-training data, post-training, and evaluation
- Run large-scale, multi-node distributed SFT and RL (GRPO) to improve multimodal and long-context reasoning (MMLongBench); current work combines GRPO with on-policy distillation and explores reward and objective design (e.g. for visual grounding)

NVIDIA ADLR, Zurich

Sep 2022 – Feb 2025

Staff Research Scientist

- First author of Eclair, a document-understanding model that extracts formatted text, tables, formulas, and reading order. Adopted across NVIDIA's pipelines for pre-training data extraction (50B+ tokens) and as a teacher for VLM distillation; shipped with open weights as Nemotron Parse
- Led the model's design: an asymmetric architecture (larger vision encoder, lightweight decoder), no positional embeddings in the decoder, multi-token prediction to reduce decoding steps, and a single prompt-controlled output format spanning text, layout, and semantic classes
- Diagnosed and fixed a data-conditioning failure where the model keyed on visual appearance rather than the text prompt, using a self-correcting labelling method (patented)
- Contributed to the token-compression design for dense OCR in Eagle 2 (NeurIPS 2025). Mentored two interns to full-time roles

Qualcomm AI Research, Amsterdam

2021 – 2022

Staff Research Scientist

- Developed hardware-efficient architectures for 4K video-to-video processing (denoising on device)
- Efficient Mixture-of-Experts architecture for conditional computation (BMVC 2022)

Qualcomm AI Research, Amsterdam

2020 – 2021

Senior Research Scientist

- Created first unsupervised approach for 3D person-localisation on video and WiFi data. Published at NeurIPS 2021 (with Max Welling) and IEEE GLOBECOM 2021 (first author)
- Video recognition via motion-augmented self-training (ICCV 2021, with Cees Snoek)
- Hand-gesture classification for mmWave data (high-frequency radio), on-device and in real-time (IEEE WCNC 2021)
- Filed 12 patent applications across positioning, gesture recognition, and efficient architectures

Microsoft, London

2018 – 2020

Senior Applied Scientist

- Applied computer vision across industry problems: self- and fully-supervised segmentation (medical and seismic imaging), object detection and counting, and real-time video action recognition
- Synthetic-data pipelines using inductive biases (low-frequency and shape signals) to improve generalisation

Microsoft, London

2016 – 2018

Applied Scientist

- Collaborated with MSR on Z3 theorem-prover for plane-gate allocations (Dubai Airport)
- Created DeepLearningFrameworks: benchmarking deep learning frameworks across single-node and multi-node on Azure. 1,700+ GitHub stars, contributions from framework creators

Charles River Associates, London

2014 – 2016

Senior Associate

- Econometric analysis of cross-sectional, time-series, and panel data for antitrust litigation
- Network/graph theory for store-customer relationships; computational geometry for merger analysis

International Growth Centre (Oxford)

2013 – 2014

Research Economist

- Research centre directed by Prof. Paul Collier. Contributed to research on firm capabilities, entrepreneurship, and trade

London School of Economics

2013 – 2014

Research Assistant to Prof. Frank Cowell

- Simulations on measures of inequality (Matlab, Stata). Causal inference (diff-in-diff) on European identity and redistribution preferences, resulting in published paper (Costa-Font & Cowell, 2015)

Publications

I Karmanov, A Deshmukh, L Voegtle, P Fischer, K Chumachenko, T Roman, J Seppanen, J Parmar, J Jennings, A Tao, K Sapra. Eclair: Extracting Content and Layout with Integrated Reading Order for Documents. arXiv 2025

K Chumachenko, A Deshmukh, J Seppanen, **I Karmanov**, C Chen, L Voegtle, P Fischer, et al. Nemotron Parse 1.1. arXiv 2025

140+ authors incl. **I Karmanov**. NVIDIA Nemotron Nano V2 VL. arXiv 2025

Authors incl. **I Karmanov**. NVIDIA Nemotron 3 Nano Omni: Efficient and Open Multimodal Intelligence. arXiv 2026

199 authors incl. **I Karmanov**. Nemotron-H: A Family of Accurate and Efficient Hybrid Mamba-Transformer Models. arXiv 2025

Z Li, G Chen, S Liu, ... **I Karmanov**, L Voegtle, P Fischer, ... Z Yu. Eagle 2: Building Post-Training Data Strategies from Scratch for Frontier Vision-Language Models. NeurIPS 2025

A Royer, **I Karmanov**, A Skliar, B Ehteshami Bejnordi, T Blankevoort. Revisiting Single-gated Mixtures of Experts. BMVC 2022

FG Zanjani, **I Karmanov**, H Ackermann, D Dijkman, S Merlin, M Welling, F Porikli. Modality-Agnostic Topology Aware Localization. NeurIPS 2021

I Karmanov, FG Zanjani, S Merlin, I Kadampot, D Dijkman. WiCluster: Passive Indoor 2D/3D Positioning using WiFi without Precise Labels. IEEE GLOBECOM 2021

K Gavriluk, M Jain, **I Karmanov**, CGM Snoek. Motion-Augmented Self-Training for Video Recognition at Smaller Scale. ICCV 2021

Y Ren, J Lu, A Beletchi, Y Huang, **I Karmanov**, et al. Hand Gesture Recognition using 802.11ad mmWave. IEEE WCNC 2021

FG Zanjani, H Ackermann, D Dijkman, **I Karmanov**, et al. Deep Learning Frameworks for Weakly-Supervised Indoor Localization. NeurIPS 2021 (Competition & Demos)

J Costa-Font, F Cowell (with research contributions from **I Karmanov**). European Identity and Redistributive Preferences. CESifo Working Paper, 2015

Book

M Salvaris, D Dean, WH Tok, **I Karmanov**. Deep Learning with Azure: Building and Deploying Artificial Intelligence Solutions on the Microsoft AI Platform. Apress, 2018

Patents

12 filed US patent applications (10 granted or pending) at Qualcomm, across RF positioning, gesture recognition, and efficient architectures.

Select Talks

I Karmanov, FG Zanjani, S Merlin, I Kadampot, D Dijkman. Demonstrating Accurate RF Sensing with Qualcomm Wi-Fi Technology. MWC 2021

L Zhang, T Wu, X Xie, A Argyriou, **I Karmanov**, K Lian. Building Production-Ready Recommendation System at Scale. KDD 2019

M Gonzalez-Fierro, D Dean, M Salvaris, **I Karmanov**. Microsoft AI Transformation. IROS 2018

I Karmanov, M Salvaris, M Gonzalez-Fierro. Distributed Training of Deep Neural Networks. ODSC 2018 / Strata 2018

I Karmanov, F Boylu-Uz. Linear Programming on Azure for Plane-Gate Optimisation. MLADS 2018

Education

London School of Economics

2012 – 2013

MSc Economics

- Thesis: Optimal Control Analysis for Tax Avoidance. Modelling a firm's optimisation problem in an infinite horizon setting, introducing reputation as an accumulated asset, solved with Hamiltonian in an optimal-control setting.
Advisor: Prof. Frank Cowell

London School of Economics

2009 – 2012

BSc Economics, First Class Honours